

2 Equilibrium Derivation (Theoretical)

1). Strategic Substitutes or Complements? Derive the profit function for firm i. Verify whether the game is one of strategic substitutes or complements. How does optimal q_i respond to changes in q_j ?

Inverse Demand and Cost Functions

- **Inverse Demand Equation:** $P = \alpha_0 + \alpha_1 Q + \alpha_2 Y + \xi$
- **Cost Function:** $C_i(q_i) = (\gamma_0 + \gamma_1 W + \omega_i) \times q_i + \gamma_2 q_i^2$

Firm i's problem

At equilibrium, Firm i maximize it's profits

$$\arg \max \pi_i = \left(\alpha_0 + \alpha_1 \sum_{k=1}^N q_k + \alpha_2 Y + \xi \right) q_i - ((\gamma_0 + \gamma_1 W + \omega_i) q_i + \gamma_2 q_i^2)$$

$$\arg \max \pi_i = (\alpha_0 + \alpha_2 Y + \xi - \gamma_0 - \gamma_1 W - \omega_i) q_i + \alpha_1 q_i \sum_{k=1}^N q_k - \gamma_2 q_i^2$$

F.O.C. w.r.t. q_i is:

$$\frac{\partial \pi_i}{\partial q_i} = \alpha_0 + \alpha_2 Y + \xi - \gamma_0 - \gamma_1 W - \omega_i + \alpha_1 Q + \alpha_1 q_i - 2\gamma_2 q_i$$

$$\frac{\partial \pi_i}{\partial q_i} = (\alpha_0 + \alpha_2 Y + \xi - \gamma_0 - \gamma_1 W - \omega_i) + \alpha_1 Q + (\alpha_1 - 2\gamma_2) q_i$$

S.O.C. w.r.t. q_j is:

$$\frac{\partial^2 \pi_i}{\partial q_i \partial q_j} = \alpha_1 < 0$$

Answer: The game is one of strategic substitutes, the optimal quantity q_i is negatively correlated to change in opponents quantity q_j

2). **Equilibrium Computation** Assume for now that the market has $N = 2$ firms. Solve for the Cournot-Nash equilibrium quantities and price. For the items below, provide an intuitive interpretation. (If you want, you can derive further mathematical results to help with interpretation, but that is not needed here.)

(a) How does the equilibrium change when marginal costs decrease (i.e., when γ_0 or W decreases)

$$\frac{\partial \pi_i}{\partial q_i} = (\alpha_0 + \alpha_2 Y + \xi - \gamma_0 - \gamma_1 W - \omega_i) + \alpha_1 Q + (\alpha_1 - 2\gamma_2) q_i = 0$$

$$\alpha_0 + \alpha_2 Y + \xi + \alpha_1 Q + \alpha_1 q = \gamma_0 + \gamma_1 W + 2\gamma_2 q + \omega_i$$

At equilibrium ($q_1 = q_2 = q^*$):

$$\alpha_0 + \alpha_2 Y + \xi + \alpha_1 (2q^*) + \alpha_1 q^* = \gamma_0 + \gamma_1 W + 2\gamma_2 q^* + \omega_i$$

$$q^* = \frac{\gamma_0 + \gamma_1 W - \alpha_0 - \alpha_2 Y - \xi}{3\alpha_1 - 2\gamma_2}$$

Given $\alpha_1 < 0$ and $\gamma_2 > 0$, then $3\alpha_1 - 2\gamma_2 < 0$. By assumptions, q^* is a non-negative quantity, thus $\gamma_0 + \gamma_1 W - \alpha_0 - \alpha_2 Y - \xi$ must be negative. Implying that $q^* = \frac{\alpha_0 + \alpha_2 Y + \xi - \gamma_0 - \gamma_1 W}{2\gamma_2 - 3\alpha_1}$ alternatively

Answer: A decrease in marginal costs (lower γ_0 or W):

1. **Firms Produce More:** Lower marginal costs reduce the cost of producing each additional unit. Firms optimally increase output ($q^* \uparrow$).
2. **Total Output Rises:** $Q = 2q^* \uparrow$, intensifying competition.
3. **Price Falls:** The increase in total output drives down the market price ($P \downarrow$) due to the inverse demand curve ($\alpha_1 < 0$).

$$P^* = \alpha_0 + 2\alpha_1 q^* + \alpha_2 Y + \xi$$

(b) How does the equilibrium change when market demand increases (i.e., when α_0 or Y increases)?

Answer: An increase in market demand (higher α_0 or Y):

1. **Price Rises:** The increase in demand shifters drives up the market price ($P \uparrow$) due to the inverse demand curve ($\alpha_0, \alpha_2 > 0$).

3). Best-Response Functions for $N = 4$ Firms Now assume there are $N = 4$ firms in the market.

(a) Write down the system of best-response functions for all firms (i.e., derive the first-order conditions for each firm).

For firm i :

$$\pi_i = (\alpha_0 + \alpha_1 Q + \alpha_2 Y + \xi) q_i - [(\gamma_0 + \gamma_1 W + \omega_i) q_i + \gamma_2 q_i^2],$$

where $Q = \sum_{j=1}^4 q_j$.

F.O.C for firm = 1,2,3,4

$$\frac{\partial \pi_i}{\partial q_i} = \alpha_0 + \alpha_1 Q + \alpha_2 Y + \xi + \alpha_1 q_i - (\gamma_0 + \gamma_1 W + \omega_i) - 2\gamma_2 q_i = 0.$$

Substitute $Q = q_i + \sum_{j \neq i} q_j$:

$$\alpha_0 + \alpha_1 \left(q_i + \sum_{j \neq i} q_j \right) + \alpha_2 Y + \xi + \alpha_1 q_i = \gamma_0 + \gamma_1 W + \omega_i + 2\gamma_2 q_i.$$

Simplify and solve for q_i :

$$q_i = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1 \sum_{j \neq i} q_j}{2(\alpha_1 - \gamma_2)}.$$

Due to symmetry:

$$q_i = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1 \sum_{j \neq i} q_j}{2(\alpha_1 - \gamma_2)}, \text{ for } i = 1, 2, 3, 4$$

(b) Explain how one could numerically solve this system to find the Cournot-Nash equilibrium.

For each firm, express their best response function in terms of their opponent's best response functions, and treat all q_i and q_j as the equilibrium quantity q^* .

For example, start with firm 1's best response function gives

$$q_1 = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1(q_2 + q_3 + q_4)}{2(\alpha_1 - \gamma_2)}, \text{ for } i = 1, 2, 3, 4$$

And:

$$q_2 = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1 \sum_{j \neq 2} q_j}{2(\alpha_1 - \gamma_2)}.$$

$$q_3 = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1 \sum_{j \neq 3} q_j}{2(\alpha_1 - \gamma_2)}.$$

$$q_4 = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1 \sum_{j \neq 4} q_j}{2(\alpha_1 - \gamma_2)}.$$

Finally, solving with substitution.

To align with my python script, where I solved the equilibrium with matrix operation, I will provide the matrix operation intuition below

Step 1: Rewrite firm i's best response as a linear system:

$$2(\alpha_1 - \gamma_2)q_i + \alpha_1 \sum_{j \neq i} q_j = \gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi.$$

Step 2: Construct a matrix form

Let $D = 2(\alpha_1 - \gamma_2)$ and $C_i = \gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi$. The system becomes:

$$Dq_1 + \alpha_1(q_2 + q_3 + q_4) = C_1,$$

$$Dq_2 + \alpha_1(q_1 + q_3 + q_4) = C_2,$$

$$Dq_3 + \alpha_1(q_1 + q_2 + q_4) = C_3,$$

$$Dq_4 + \alpha_1(q_1 + q_2 + q_3) = C_4.$$

This can be written as $\mathbf{A}\mathbf{q} = \mathbf{C}$, where:

$$\mathbf{A} = \begin{bmatrix} D & \alpha_1 & \alpha_1 & \alpha_1 \\ \alpha_1 & D & \alpha_1 & \alpha_1 \\ \alpha_1 & \alpha_1 & D & \alpha_1 \\ \alpha_1 & \alpha_1 & \alpha_1 & D \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix}.$$

Step 3: Solve the system numerically

$$\mathbf{q} = \mathbf{A}^{-1}\mathbf{C}.$$

3. Generate Fake Data

Refer to the python file

4. Estimation

Market Power and Concentration

(1) Regress market-level market power on HHI. What do you find?

Dep. Variable:	L_m	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.001
Method:	Least Squares	F-statistic:	0.07776
Date:	Mon, 03 Mar 2025	Prob (F-statistic):	0.780
Time:	02:18:40	Log-Likelihood:	3757.4
No. Observations:	1000	AIC:	-7511.
Df Residuals:	998	BIC:	-7501.
Df Model:	1	Covariance Type:	nonrobust

	coef	std err	t	P< t	[0.025	0.975]
const	0.7155	0.954	0.750	0.454	-1.157	2.588
HHI_m	-1.0642	3.816	-0.279	0.780	-8.553	6.425

Omnibus:	0.690	Durbin-Watson:	1.978
Prob(Omnibus):	0.708	Jarque-Bera (JB):	0.637
Skew:	-0.061	Prob(JB):	0.727
Kurtosis:	3.021	Cond. No.:	2.27e+04

Table 1: OLS Regression Results of Market Power on HHI

(2) Discuss briefly the problems involved in estimating this regression.

The regression result suggests that the Herfindahl-Hirschman Index doesn't explain the variation in the market power index, despite they are closely related theoretically. The estimate has an unexpected sign (they should be positively correlated) and is not statistically significant.

$$L^{market} = \sum_{i=1}^N s_i L_i$$

$$HHI = \sum_{i=1}^N s_i^2$$

This is likely due to omitted variable bias, where key explanatory variable(s) that are correlated with the market share (s_i) are not included. For example, demand/cost shifters.

Demand and Supply Estimation

(1) Estimate both demand and marginal cost parameters using OLS

The Estimation Strategy:

The F.O.C implies:

$$MC_i = MR_i$$

Step 1: Estimate Demand Step 2: Extract the $\hat{\alpha}_1$ from the demand OLS

Step 3: Construct $\hat{MC}_i$ using $\hat{\alpha}_1$

$$MC_i = MR_i = P_t + \frac{\partial P_t}{\partial Q_t} q_{it}$$

Step 4: Regress $\hat{MC}_i$ on the rest of the variables

$$\hat{MC}_i = \gamma_0 + \gamma_1 W_t + 2\gamma_2 q_{it} + \omega_{it}$$

The Estimation Results:

Table 2: Demand OLS Regression Results

Dep. Variable:	Market Price	R-squared:	0.428
Model:	OLS	Adj. R-squared:	0.427
Method:	Least Squares	F-statistic:	372.9
Date:	Sun, 02 Mar 2025	Prob (F-statistic):	1.25e-121
Time:	17:36:45	Log-Likelihood:	-844.31
No. Observations:	1000	AIC:	1695.
Df Residuals:	997	BIC:	1709.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	38.8956	1.645	23.638	0.000	35.667	42.125
Market Quantity	-0.0385	0.043	-0.887	0.375	-0.124	0.047
Y	0.5252	0.028	18.512	0.000	0.470	0.581

¹ The condition number is large, 6.93e+03. This might indicate that there are strong multicollinearity or other numerical problems.

² Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 3: Marginal Cost OLS Regression Results

Dep. Variable:	Firm Marginal Price	R-squared:	0.320
Model:	OLS	Adj. R-squared:	0.320
Method:	Least Squares	F-statistic:	940.1
Date:	Sun, 02 Mar 2025	Prob (F-statistic):	0.00
Time:	17:48:15	Log-Likelihood:	-3707.8
No. Observations:	4000	AIC:	7422.
Df Residuals:	3997	BIC:	7440.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	47.0869	0.525	89.741	0.000	46.058	48.116
Cost Shifter	1.4727	0.039	38.049	0.000	1.397	1.549
Firm Quantity	0.9921	0.037	26.892	0.000	0.920	1.064

¹ Standard Errors assume that the covariance matrix of the errors is correctly specified.

(2) Estimate both demand and marginal cost parameters using 2SLS. Justify the choice of the instrumental variables.

The Estimation Strategy:

The strategy is pretty similar to the OLS base case. I run regressions for both ordinary demand/supply and inverse demand and supply, because I wasn't sure if instrumenting Q instead of P is okay in the inverse demand/supply case. The arguments for IVs choice is exactly the same as in problem set 1. Here I would just paste the old one.

IV for The Demand Equation: W (Cost Shifter)

- **Relevance:**

$$\text{Corr}[W, P] \neq 0$$

$$\text{Corr}[W, Q] \neq 0$$

- W affects **price (P)** through the **Supply Side** (simultaneous equations). Higher input costs reduce supply (Q_S) and vice versa, resulting in a change in the equilibrium price **price (P*)**.

- **Exclusion Restriction:**

- W is a **Cost Shifter**, thus it doesn't affect quantity demand or the price demand. It only affect the demand side variables through the supply.

- **Exogeneity:**

$$\text{Corr}(W, \xi) = 0$$

- W is an exogenous variable that is related to the supply side only, so it is uncorrelated with the unobserved demand shock ξ .

Conclusion:

W is a valid instrument for P_D, Q_D .

IV for The Supply Equation: Y Demand Shifter

- **Relevance:**

$$\text{Corr}[Y, P] \neq 0$$

$$\text{Corr}[Y, Q] \neq 0$$

- Y affects **price (P)** through **Demand Side** (simultaneous equations). Higher input income increases demand (Q_D) and vice versa, resulting in a change in the equilibrium price **price (P*)**.

- **Exclusion Restriction:**

- Y is a **Demand Shifter**, thus it does *not* directly affect the quantity supplied or the price supplied. It only affect the supply side variables through the demand.

- **Exogeneity:**

$$\text{Corr}[Y, \omega] = 0$$

- Y is an exogenous variable that is related to the demand side only, so it is uncorrelated with the supply shock ω .

Conclusion:

Y is a valid instrument for MC_S, Q_S .

The Estimation Results for Demand:

Table 4: 2SLS Ordinary Demand Estimation Summary

Dep. Variable:	Market Quantity	R-squared:	0.3642	Adj. R-squared:	0.3629
Estimator:	IV-2SLS	F-statistic:	1008.0	P-value (F-stat):	0.0000
No. Observations:	1000	Distribution:	chi2(2)	Cov. Estimator:	robust
Date:	Sun, Mar 02 2025	Time:	18:38:22		
	Coef.	Std. Err.	t	P-value	95% CI
Intercept	51.289	1.9782	25.928	0.0000	[47.412, 55.166]
Demand Shifter	0.7628	0.0302	25.250	0.0000	[0.7036, 0.8220]
Market Price	-0.5302	0.0488	-10.866	0.0000	[-0.6258, -0.4346]
Endogenous: Market Price					
Instruments: Cost Shifter					
Robust Covariance (Heteroskedastic)					
Debiased: False					

Table 5: 2SLS Inverse Demand Estimation Summary

Dep. Variable:	Market Price	R-squared:	-0.6110	Adj. R-squared:	-0.6142
Estimator:	IV-2SLS	F-statistic:	405.16	P-value (F-stat):	0.0000
No. Observations:	1000	Distribution:	chi2(2)	Cov. Estimator:	robust
Date:	Sun, Mar 02 2025	Time:	18:48:08		
	Coef.	Std. Err.	t	P-value	95% CI
Intercept	96.733	5.7131	16.932	0.0000	[85.536, 107.93]
Demand Shifter	1.4387	0.0899	16.008	0.0000	[1.2625, 1.6148]
Market Quantity	-1.8861	0.1736	-10.866	0.0000	[-2.2263, -1.5458]
Endogenous: Market Quantity					
Instruments: Cost Shifter					
Robust Covariance (Heteroskedastic)					
Debiased: False					

Table 6: Comparison of True Demand Parameters and Estimates

Model	True Value	Estimate	β	Estimate Derivation
Ordinary Demand (Table 3)	α_0 : 100	96.733	51.298	$\alpha_0 = -\beta_0\alpha_1$
	α_1 : -2	-1.886	0.7628	$\alpha_1 = \frac{1}{\beta_1}$
	α_2 : 1.5	1.439	-0.5302	$\alpha_2 = -\beta_2\alpha_1$
Inverse Demand (Table 4)	α_0 : 100	96.733	-	-
	α_1 : -2	-1.886	-	-
	α_2 : 1.5	1.439	-	-

Notes: Ordinary Demand estimates ($\alpha_0, \alpha_1, \alpha_2$) are derived based on column 4 Estimation Derivation with the regression output ($\beta_0, \beta_1, \beta_2$) as shown in Table 3. Inverse Demand beta estimates ($\alpha_0, \alpha_1, \alpha_2$) are directly from the regression output as shown in table 4.

The Estimation Results for Marginal Cost (MC):

Table 7: 2SLS Ordinary MC (Supply) Estimation Summary

Dep. Variable:	Firm Marginal Cost	R-squared:	-1.9106	Adj. R-squared:	-1.9120
Estimator:	IV-2SLS	F-statistic:	315.78	P-value (F-stat):	0.0000
No. Observations:	4000	Distribution:	chi2(2)	Cov. Estimator:	robust
Date:	Sun, Mar 02 2025	Time:	19:59:30		
	Parameter	Std. Err.	T-stat	P-value	95% CI
Intercept	-1.4291	0.9341	-1.5298	0.1261	[-3.2600, 0.4018]
Cost Shifter	-0.9013	0.0519	-17.353	0.0000	[-1.0031, -0.7995]
Firm Quantity	0.4466	0.0267	16.709	0.0000	[0.3942, 0.4990]
Endogenous: Firm Quantity					
Instruments: Demand Shifter					
Robust Covariance (Heteroskedastic)					
Debiased: False					

Table 8: 2SLS Inverse MC (Supply) Estimation Summary

Dep. Variable:	Firm Quantity	R-squared:	-0.7388	Adj. R-squared:	-0.7396
Estimator:	IV-2SLS	F-statistic:	975.87	P-value (F-stat):	0.0000
No. Observations:	4000	Distribution:	chi2(2)	Cov. Estimator:	robust
Date:	Sun, Mar 02 2025	Time:	20:05:03		
	Parameter	Std. Err.	T-stat	P-value	95% CI
Intercept	3.1998	1.9002	1.6839	0.0922	[-0.5245, 6.9241]
Cost Shifter	2.0181	0.0662	30.506	0.0000	[1.8885, 2.1478]
Firm Marginal Cost	2.2391	0.1340	16.709	0.0000	[1.9764, 2.5017]
Endogenous: Firm Marginal Cost					
Instruments: Demand Shifter					
Robust Covariance (Heteroskedastic)					
Debiased: False					

Table 9: Comparison of True Supply Parameters and Estimates

Model	True Value	Estimate	β	Estimate Derivation
Ordinary MC (Table 6)	γ_0 : 5	3.200	-1.429	$\gamma_0 = -2\beta_0\gamma_2$
	γ_1 : 2	2.018	-0.901	$\gamma_1 = -2\beta_1\gamma_2$
	γ_2 : 1	1.120	0.447	$\gamma_2 = \frac{1}{2\beta_2}$
Inverse MC (Table 7)	γ_0 : 5	3.200	-	-
	γ_1 : 2	2.018	-	-
	γ_2 : 1	1.120	2.2391	$\gamma_2 = \frac{\beta_2}{2}$

Notes: - Ordinary Demand estimates ($\gamma_0, \gamma_1, \gamma_2$) are derived based on column 4 Estimation Derivation with the regression output ($\beta_0, \beta_1, \beta_2$) as shown in Table 6 - Inverse Demand beta estimates ($\gamma_0, \gamma_1, \gamma_2$) are directly from the regression output as shown in table 7, except for γ_2 which is derived with the regression output β_2

(3) Compare the OLS and 2SLS estimates. Interpret the differences in the results.

I will answer this question in two-folds. Firstly I want to discuss the results from the inverse and the ordinary estimations. Then I will move to compare the 2SLS and OLS results.

For simultaneous equations, both the original or the inverse equation estimation gives the correct estimates, as evidences shown in Table 5 and 7. The results confirm my thoughts earlier on the simultaneous equation, that the endogeneity exists systemically in both Quantity and Price and instrumenting either one address the issue.

As shown in Table 9, 2SLS gives closer estimates to the demand true parameter values. All estimate values are within 1 standard deviation to the true parameter values (in the bracket below each estimates).

Table 10: Comparison of OLS and 2SLS Demand Estimates

Parameter	OLS Estimate	2SLS Estimate	True Value
α_0	38.900 (1.645)	96.733 (5.713)	100 -
α_1	-0.039 (0.043)	-1.886 (0.0899)	-2 -
α_2	0.525 (0.028)	1.434 (0.174)	1.5 -

As shown in Table 10, 2SLS gives closer estimate to the marginal cost true parameter values. All estimate values are within 1 standard deviation to the true parameter values (in the bracket below each estimates).

Table 11: Comparison of OLS and 2SLS Supply Estimates

Parameter	OLS Estimate	2SLS Estimate	True Value
γ_0	47.0869 (0.525)	3.1998 (1.900)	5 -
γ_1	1.4727 (0.039)	2.0181 (0.066)	2 -
γ_2	0.9921 (0.037)	2.2391 (0.134)	1 -

Conclusion: OLS estimates are biased due to failure to account for the endogeneities in simultaneous equations. The issue is addressed with 2SLS estimations which give better estimates that are within 1 standard deviations to the true value.

5. Counterfactual Analysis

Theoretical Analysis

(1) Explain intuitively what the direct effect of this cost reduction on firm i's quantity is (i.e., holding other firms' quantities fixed). (No math needed here.)

The Cost Function:

$$C_i(q_i) = (\gamma_0 + \gamma_1 W + \omega_i) \times q_i + \gamma_2 q_i^2$$

The Marginal Cost Function:

$$MC_i(q_i) = (\gamma_0 + \gamma_1 W + \omega_i) + 2\gamma_2 q_i$$

$$MC_i(q_i) = \gamma_0 + \gamma_1 W + 2\gamma_2 q_i + \omega_i$$

As shown above, the marginal cost is constant in γ_0 . γ_0 captures the baseline production cost that is constant in the margin (eg. raw materials). Thus, when holding everything else equal, a reduction in the fixed per-unit cost γ_0 should incentivize more production, and quantity produced increases as an result

(2) Firm j also changes its quantities in response to reduced marginal costs. Explain intuitively what the indirect, strategic effect of q_j on q_i is. Can we predict if q_i increases or decreases taking both direct and indirect effects? (Again, no math needed here.)

Direct Effect: When firms react to a shock, they refer to the best responses as if everything else were held constant. Naturally, all firms would come to the same conclusion that it is in their best interest to increase the quantity produced as a direct effect of cost reduction. Firm j would do the same.

Indirect Strategic Effect: At a macro level, all firms (including firm j 's) face lower costs, and all increase production. This raises total market output, lowering the market price (due to the downward sloping demand curve). The price drop reduces the profitability of additional production for firm i .

The Net Impact: The direct effect channeling in from the best response function, leading firm i to increase q_i . The strategic effect partially offsets the direct gain but does not reverse it. As a result, the Firm i 's equilibrium quantity increases, though less than it would if given the same shock in a monopoly setting.

Numerical Analysis

(1) Derive the new equilibrium outcomes when marginal costs decrease. Consider the average market and use your preferred estimates for the parameter values. What happens to q_{im}^N , Q_m and P_m Interpret the results.

Variable	Old Equilibrium Average	New Equilibrium Average	Static BR value
q1	14.017070	14.154173	128.889515
q2	14.008215	14.145318	128.864417
q3	14.014074	14.151177	128.881023
q4	14.013631	14.150735	128.879769
Q_m	56.052991	56.601404	-
P_m	62.992041	61.957705	-

Table 12: Comparison of Old/New Equilibrium Averages and Static Best Response

The result is in align with the theoretic analysis above. All firms are incentivize to produce more, which is less than the change in price would have happened in the monopoly setting. Precisely, if holding everything else constant the firm would adjust the production as described in the following best response function. Table 12 the last column (Static BR Value) demonstrates the changes would have been observed if assuming for a holding all other factors constant.

$$q_i = \frac{\gamma_0 + \gamma_1 W + \omega_i - \alpha_0 - \alpha_2 Y - \xi - \alpha_1 \sum_{j \neq i} q_j}{2(\alpha_1 - \gamma_2)}, \text{ for } i = 1, 2, 3, 4$$

Comparing to a static game, the individual firm's production in a simultaneous multi-player's game, realize less change in quantity when a shock happens.